

NMEC595 - Research Methodology

Lecture 7: Processing and Analysis of Data

(Reference: Ch-7, Research Methodology, C.R. Kothari, 2nd Edition)

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Example Dataset: Google Transparency Report

In such a tabular dataset columns represent features/variables and rows represent observations/instances.

country	cr_req	cr_comply	ud_req	ud_comply	hemisphere	hdi
Argentina	21	100	134	32	southern	very high
Australia	10	40	361	73	southern	very high
Belgium	<10	100	90	67	northern	very high
Brazil	224	67	703	82	southern	high
...	...	...	...	...	...	...
United States	92	63	5950	93	northern	very high

Source: Google Transparency Report (2010-2011)

Variable Types in the Dataset

Column	Variable Type
country	Categorical (Nominal) Names without inherent order
cr_req	Numerical (Discrete) Count data (whole numbers)
cr_comply	Numerical (Continuous) Percentage values (0–100)
ud_req	Numerical (Discrete) Count data (whole numbers)
ud_comply	Numerical (Continuous) Percentage values (0–100)
hemisphere	Categorical (Nominal) Northern or Southern
hdi	Categorical (Ordinal) Ranked levels (high, very high, etc.)

Data Processing Can be Challenging

A mechanical engineering lab runs **500 heat-treatment trials** on alloy-steel test specimens to study process–property trends:

- Raw data: 500 trial records with temperatures (400°C–900°C) and measured outputs (e.g., UTS, hardness)
- Some entries missing (sensor dropout or incomplete log sheets)
- Data scattered across sources (furnace log, tensile tester export, operator notes)
- **How do we convert these raw records into analyzable information?**

This is where data processing becomes essential!

- **Processing:** Organizing raw records into usable form
- **Analysis:** Computing measures and identifying relationships

Processing vs. Analysis

Processing (Preparation)

- Examining raw trial records
- Detecting errors and omissions
- Coding operator notes
- Classifying temperatures/strength levels
- Organizing results in tables

Analysis (Interpretation)

- Computing averages (e.g., mean UTS)
- Measuring dispersion (e.g., variance)
- Finding relationships (e.g., temper temp vs hardness)
- Testing hypotheses
- Drawing conclusions

Key Insight:

Both are **essential** for scientific research. Processing without analysis leaves data unused; analysis without proper processing produces invalid results.

P1: Raw Data Errors

Heat-Treatment Trials (same 500 trials)

Quality control team compiles measurements from tensile tests linked to selected trials:

- Technician 1 records: “250 MPa”
- Technician 2 writes: “2500 MPa” (typo - extra zero)
- Technician 3 notes: “250–” (incomplete entry)
- Technician 4 uses shorthand: “250M”

Same variable recorded 4 different ways!

Solution: EDITING - careful examination and correction

What is Editing?

Definition: Examining collected raw data to detect errors and omissions, then correcting them

Editing ensures data are:

- 1 **Accurate** - values match measurements
- 2 **Consistent** - same units and format throughout
- 3 **Complete** - missing data identified
- 4 **Uniform** - standardized recording

Types of Editing: Field vs. Central

Field Editing

- Done immediately after recording trial results
- By the investigator/technician
- Same day as the run/test
- Can clarify illegible entries
- Can check logs while context is fresh

Advantage: Information is fresh

Best Practice

Use **both**: Field editing catches obvious errors immediately, central editing ensures consistency across all 500 trials

Central Editing

- Done after all trials are completed
- By a dedicated editor/analyst
- Applies uniform standards to all trials
- Cross-checks related fields (units, ranges)
- Flags outliers for verification

Advantage: Systematic and uniform

Editing Guidelines

When editing experimental records, follow these rules:

- 1 **Be familiar** with expected ranges (e.g., UTS for the alloy)
- 2 **Mark corrections** with single line (keep original legible)
- 3 **Use distinctive color** (e.g., red pen for edits)
- 4 **Initial all changes** - who made the edit?
- 5 **Date the edit** - when was correction made?

Example:

Original entry: "2500 MPa" (suspicious value)

Edited as: ~~2500~~ 250 MPa

Marked: [JB, 29-Dec-2025, Field Edit]

What if you can't determine correct value?

- Cross-reference with tensile machine export and specimen ID
- **Strike out** and mark as "No Answer"
- Contact original recorder if possible

Editing Guidelines in the Digital Age

Why do these rules still matter today?

In modern labs, data is rarely edited on paper. Instead, the **same principles** are applied digitally.

- 1 **Know expected ranges** → Automated range checks, validation rules, and sanity filters in code
- 2 **Do not erase original values** → Raw data is kept **unchanged**; cleaned data is stored separately
- 3 **Mark every correction clearly** → Flags, masks, or new columns (e.g., UTS_raw vs UTS_clean)
- 4 **Initial all changes** → Version control (Git), logs, or metadata record **who** changed the data
- 5 **Date every edit** → Timestamps in logs, commit history, or database audit trails

Key idea:

Modern data editing is reproducible, traceable, and auditable. This is essential for machine learning, digital twins, and scientific credibility.

P2: Diverse Response Formats

Heat-Treatment Trials (same 500 trials)

Field in the trial sheet: “Observed issue after quench” (operator note)

Responses vary:

- “distortion”
- “warped”
- “bent”
- “crack”
- “surface crack”
- “ok”
- “no issue”

Same meaning written many different ways!

How do we count issues and analyze them statistically?

What is Coding?

Definition: Assigning numerals or symbols to responses so data can be organized into categories

Example Coding Scheme (Observed issue after quench):

Code	Issue Type
0	No issue
1	Distortion / warpage
2	Cracking
3	Surface oxidation / scale
4	Hardness out-of-spec
5	Logging / sensor issue
6	Other

Benefits:

- Reduces many responses to few categories
- Allows statistical analysis
- Enables computer processing

Rules for Good Coding

Every coding scheme must satisfy three properties:

1 Exhaustiveness

- **Every** possible issue has a code
- Example: “Other” captures rare notes

2 Mutual Exclusivity

- Each recorded issue fits in **only one** category
- Categories don't overlap
- Example: Don't have both “Crack” and “Surface crack” as separate categories unless clearly separated by rule

3 Unidimensionality

- Each category defined by **one** characteristic only
- Example: Don't mix “Issue Type” with “Severity”

Bad Coding: Code = “Crack (severe)” ← mixes issue and severity

Good Coding: Code = “Cracking” ← single concept

When to Code?

Best Time: During Data Sheet / Form Design

Pre-code the trial sheet:

- Include codes next to each possible issue category
- Operator selects the closest code during recording
- Reduces post-hoc interpretation errors
- Enables direct computer entry

Example of Pre-coded Field:

Observed issue after quench:

- [0] No issue
- [1] Distortion / warpage
- [2] Cracking
- [3] Surface oxidation / scale
- [4] Hardness out-of-spec
- [5] Logging / sensor issue
- [6] Other

P3: Too Much Raw Data

Heat-Treatment Trials (same 500 trials)

500 temperature measurements from the furnace log:

Original raw data: 500 scattered numbers

- 523°C, 547°C, 489°C, 612°C, 498°C, ... (continues for 500 entries)

What patterns exist? Is the process centered? Is it spread out?

We can't see patterns in 500 raw numbers!

Solution: Classification - arrange data into homogeneous groups

What is Classification?

Definition: Arranging data into groups or classes based on common characteristics

Example - Classify Austenitizing Temperature Data:

Temperature Range	Frequency
400–500°C	45
500–600°C	187
600–700°C	156
700–800°C	92
800–900°C	20
Total	500

Now we can see:

- Most trials cluster around 500–700°C
- Very few extremely high temperatures
- The process shows a clear distribution pattern

Two Ways to Group Data (Classification)

1. By Attributes (Qualitative):

- Group by category labels (not numbers)
- Example:
 - Quench medium: Water, Oil, Polymer
 - Issue code: Distortion, Cracking, None
 - Sensor status: OK, Missing, Drift
- Each sample is sorted by “name” or “type”
- Can combine categories for more detail (e.g., “Oil” + “Cracking”)

2. By Class Intervals (Quantitative):

- Group by number ranges
- Example:
 - Austenitize temp: 400–450°C, 450–500°C, etc.
 - UTS: 700–800 MPa, 800–900 MPa, etc.
 - Hardness: 30–35 HRC, 35–40 HRC, etc.
- Each sample is sorted by “range” (not just its exact value)
- Used for making frequency tables and histograms

Setting Up Class Intervals

Problem For 500 austenitizing temperature measurements (400–900°C), how many classes should we create?

Guidelines:

- ❶ **Too few classes** (e.g., 2–3): Details lost
 - “400–900°C” loses pattern information
- ❷ **Too many classes** (e.g., 50+): Data scattered, patterns unclear
 - “400–410, 410–420, ...” creates mostly empty classes
- ❸ **Optimal:** 5–15 classes (typical recommendation)

Sturges Formula (optional guideline):

$$i = \frac{\text{Range}}{1 + 3.3 \log(N)}$$

- i = class interval size
- Range = Max - Min
- N = number of observations

Sturges Formula - Complete Worked Example

Real Engineering Problem (same 500 trials):

You have 500 austenitizing temperature measurements from heat treatment:

- Minimum temperature: 400°C
- Maximum temperature: 900°C
- Question: **How many classes and what interval width?**

Step-by-Step Calculation:

- 1 Calculate Range: $R = 900 - 400 = 500^\circ\text{C}$
- 2 Identify sample size: $N = 500$ measurements
- 3 Apply Sturges Formula:

$$i = \frac{R}{1 + 3.3 \log(N)} = \frac{500}{1 + 3.3 \log(500)}$$

- 4 Calculate logarithm: $\log(500) \approx 2.699$

Sturges Formula

Continuing from previous calculation...

- 5 Substitute values:

$$i = \frac{500}{1 + 3.3(2.699)} = \frac{500}{1 + 8.907} = \frac{500}{9.907} \approx 50.5^{\circ}\text{C}$$

- 6 Round to “nice” multiple: Choose $i = 50^{\circ}\text{C}$ (multiple of 10)

- 7 Calculate number of classes: $k = \frac{R}{i} = \frac{500}{50} = 10$ classes

Final Result - Create These Classes:

Class #	Temperature Range	Class Width
1	400–450°C	50°C
2	450–500°C	50°C
3	500–550°C	50°C
...	...	...
10	850–900°C	50°C

Choosing Class Limits

How to select upper and lower boundaries?

Key Principle: Choose class limits at “nice” numbers (multiples of 2, 5, 10, etc.)

Example - Temperature Data (same trials):

Exclusive Type	Inclusive Type
400–450	400–449
450–500	450–499
500–550	500–549
550–600	550–599

Choosing Class Limits

Exclusive Type (Half-open):

Upper limit is **not** included in the class

- Example: 400–450 means $400 \leq x < 450$
- Value 450°C goes in 450–500, not 400–450
- Common for continuous measurements

Inclusive Type:

Both limits are included in each class (often for integer data)

- Example: 400–449, 450–499, 500–549, ...
- Value 450°C goes in 450–499 (no overlap)
- If data has decimals, use boundaries like 399.5–449.5, 449.5–499.5, ...

In practice: Engineering measurements are often treated with exclusive classes to avoid overlap.

The Problem: Organized But Scattered

Current Situation

For the same 500 heat-treatment trials, we now have:

- Edited data (errors corrected)
- Coded data (operator notes in numerical form)
- Classified data (temperatures organized into 10 classes)
- Frequency counts (how many trials per class)

But this information is scattered across logs, sheets, and notes.

How do we present this clearly to others?

Solution: TABULATION - organize data into formatted tables

What is Tabulation?

Definition: Arranging processed data in a compact table form suitable for analysis and presentation

Example - Frequency Distribution Table (Austenitizing Temp, 500 trials):

Temperature Range (°C)	Frequency
400–450	12
450–500	33
500–550	90
550–600	97
600–650	80
650–700	76
700–750	50
750–800	42
800–850	12
850–900	8
Total	500

Why Tabulation Matters

Benefits of tabulation:

- 1 **Conserves space** - 500 values become 1 table
- 2 **Facilitates comparison** - easy to see dominant ranges
- 3 **Enables calculations** - mean, variance, etc. from the table
- 4 **Detects errors** - total should match N ; discrepancies = problems
- 5 **Communicates clearly** - stakeholders interpret quickly

Example Insight (same trials):

- Many trials fall in 500–700°C ($90 + 97 + 80 + 76 = 343$ of 500)
- Few trials occur above 800°C ($12 + 8 = 20$ of 500)
- **This guides process centering and control!**

Principles of Good Tables

When creating tables, follow these rules:

- 1 **Clear, concise title** - placed above table
- 2 **Distinct number** - for reference in text (Table 1, Table 2, etc.)
- 3 **Clear column headings** - brief but descriptive
- 4 **Units of measurement** - explicitly stated ($^{\circ}\text{C}$, MPa, min, etc.)
- 5 **Totals** - usually in bottom row and rightmost column
- 6 **Footnotes** - explanations placed below table
- 7 **Source citation** - where data came from
- 8 **Alignment** - decimal points aligned vertically
- 9 **Avoid abbreviations** - unless essential (spell out first use)
- 10 **Logical arrangement** - by magnitude, time, geography (as appropriate)

Bad Example: Random numbers with no labels

Good Example: The frequency table for the 500 heat-treatment trials

Extra: File Naming for Multi-Parameter and Time-Based Studies

Goal: Make every file **self-explanatory**, searchable, and safe to use in automation.

Core rules:

- Include all **varying parameters** in names (e.g., temperature, batch, run).
- Use a **fixed order and format** so files sort correctly.
- Add **time or date** for repeated runs or long studies.
- Avoid spaces and special characters; use _ or -.
- Never overwrite files—each run gets a unique name.

Why this matters:

- Enables scripting and ML pipelines.
- Prevents data loss and confusion.
- Makes results reproducible months or years later.

Example: Practical Naming Pattern

Recommended pattern:

YYYY-MM-DD_param1_param2_runXX

Example (heat-treatment study):

- 2025-02-06_temp700_quench-oil_run03/
- 2025-02-06_temp750_quench-water_run01/

Inside each folder:

- raw/data.csv (original measurements)
- processed/data_clean.csv
- results/summary.json
- metadata.yaml (all parameters + notes)

Key idea: Names identify the run; metadata explains it in detail.

Summary: Complete Processing Flow

Four Essential Steps (same 500 heat-treatment trials):

1 EDITING

- Detect and correct errors (units, typos, missing fields)
- Ensure consistency and completeness across all trial records

2 CODING

- Assign codes to operator notes (e.g., distortion, cracking, none)
- Standardize diverse textual formats

3 CLASSIFICATION

- Arrange numerical variables into groups (e.g., temperature classes)
- Use class intervals for continuous data
- Apply Sturges formula as a guideline

4 TABULATION

- Organize results into formal tables
- Present clearly for analysis and reporting

Next: Tabulation and Statistical Analysis